

A Hybrid Neuro-Symbolic-Probabilistic Framework: Combining a Trajectory Foundation Model, a Bayesian Belief Network, and an LLM Agent for Next-Location Prediction on GPS Trajectory Data

1 Introduction

The proliferation of satellite positioning (GPS) devices and location-based services has generated a massive volume of trajectory data, recording spatio-temporal point sequences $T = \{(p_1, t_1), (p_2, t_2), \dots, (p_n, t_n)\}$ that describe the movement of people and vehicles [1]. Mining GPS trajectory data plays a central role in numerous real-world applications: urban planning, intelligent transportation systems, point-of-interest (POI) recommendation, and, in particular, the task of *next-location prediction* – one of the core problems in this field.

Over roughly the past decade, deep learning methods (RNNs, Transformers, spatio-temporal graphs) have become the de facto standard for next-location prediction, achieving high quantitative accuracy by capturing latent statistical patterns in historical data [2]. However, these models typically behave as "black boxes," lack interpretability, and struggle to leverage rich contextual knowledge (e.g., the functional structure of an area, cultural habits, special events) that is not explicitly represented in raw numerical data.

The emergence of large language models (LLMs) has opened up a new line of research: using an LLM directly as a reasoning engine over trajectory data, exploiting the broad world knowledge LLMs acquire from large-scale text corpora [3, 4]. Agentic frameworks such as AgentMove [3] have shown that LLMs can achieve competitive zero-shot performance by decomposing the task into specialized reasoning modules (personal memory, world knowledge, collective knowledge). However, recent systematic evaluations – notably CityBench [6] and Traj-MLLM [7] – have identified an inherent limitation: LLMs perform well on tasks that require semantic reasoning and common sense, but *fail markedly* on tasks that require numerical competence and precise quantitative prediction, such as predicting specific geographic coordinates.

This raises a central research question: *can the quantitative accuracy of specialized deep learning models be combined with the semantic reasoning ability of LLMs, in a way that simultaneously improves performance, reduces computational cost, and yields interpretable predictions?* A number of recent works have approached this problem from different angles: NextLocLLM [8] fuses semantic representation and coordinate regression within a *single* fine-tuned LLM backbone; TrajAgent [9] uses an LLM as an *orchestrator* that selects a specialized model for each task. However, no existing work exploits a design with *separate inference stages at prediction time*, in which a lightweight quantitative model produces calibrated-probability candidates and an LLM agent supplies semantic evidence and world knowledge – a design that clearly separates the "quantitative" and "reasoning" roles while substantially shrinking the search space the LLM has to process.

A second issue, largely overlooked by recent LLM-based work, concerns *how* to fuse different sources of evidence (the quantitative output of a deep model, an assessment of personal habits, an assessment of urban context) into a final decision. Most existing work (AgentMove, LLM-Mob, LLM-ZS) delegates this fusion step entirely to the LLM via free-form chain-of-thought text – a

mechanism that lacks explicit probabilistic semantics, is difficult to calibrate, and whose generated explanation may not faithfully reflect the actual decision process (*unfaithful explanation*), a limitation widely documented in the chain-of-thought literature. Meanwhile, an older line of work has shown that Bayesian networks are an effective tool for modeling human trajectories as a sequence of latent variables (location, time of day, day of week), supporting real-time statistical updating without retraining the entire model, unlike deep-learning-based approaches [12]. This suggests a natural complementary direction: *instead of letting the LLM freely re-rank candidates via text, one can use the LLM to extract qualitative evidence (personal-habit fit, semantic fit) and then fuse this evidence with the CGM’s quantitative probability through an explicit Bayesian Belief Network (BBN)*, yielding a theoretically calibrated posterior and a *faithful* explanation derived directly from the network’s decomposition, rather than from free-form text generated by the LLM.

Contributions. This paper proposes a hybrid *neuro-symbolic-probabilistic* framework for next-location prediction on GPS trajectory data, with the following main contributions:

1. We propose a three-stage architecture: (i) a lightweight quantitative model that produces top- k location candidates with calibrated probabilities; (ii) an LLM agent that extracts semantic evidence (rather than directly re-ranking) from personal memory and world knowledge; (iii) a Bayesian belief network that fuses this evidence with the quantitative probability to derive the final posterior.
2. We design a lightweight (naive-Bayes-style) Bayesian network that treats the CGM’s calibrated probability as the prior, and treats the LLM’s assessments (degree of fit with personal habits, degree of fit with urban context) as conditionally independent likelihoods, enabling incremental Bayesian updating as new evidence arrives, without retraining.
3. We propose a faithful-explanation mechanism: instead of extracting a rationale from the LLM’s free-form text, the explanation is derived directly from the log-probability contribution of each evidence source in the Bayes formula, guaranteeing that the explanation reflects the actual computation that produced the prediction.
4. We design a probability calibration mechanism at both the CGM layer and the LLM-evidence layer, integrated into the inference pipeline to reduce the number of tokens processed compared to LLM-only approaches.
5. We conduct a comprehensive empirical evaluation across multiple cities, comparing against purely quantitative baselines, LLM-only baselines, and classical Bayesian networks, while additionally measuring inference cost, calibration quality (ECE), and explanation faithfulness – aspects that have received little attention in related work.

The rest of the paper is organized as follows: Section 2 surveys related work; Section 3 presents the proposed method in detail; Section 4 presents the experimental design (with result placeholders to be filled in after running the experiments).

2 Related Work

2.1 Deep learning and foundation models for trajectory data

The pre-LLM era saw substantial progress in trajectory representation learning based on RNNs, Transformers, and contrastive learning [2]. A more recent trend is the shift toward *trajectory foundation models*, pretrained in a self-supervised manner on large-scale data to produce general-purpose representations that can be fine-tuned for a variety of downstream tasks. Representative examples include UniTraj [10], pretrained on 2.45 million worldwide trajectories (the WorldTrace

dataset) using an encoder–decoder architecture, and PMT [11], an autoregressive Transformer trained on nearly 42 million location sequences. A common trait of this group of methods is high quantitative accuracy, but a lack of ability to leverage unstructured contextual knowledge and to generate explanations for predictions.

2.2 LLMs as direct reasoners for mobility prediction

The second group of methods uses an LLM directly as the predictor, exploiting its zero-shot reasoning ability. LLM-Mob [4] and LLM-ZS [5] were early efforts demonstrating that LLMs can predict the next location without retraining. AgentMove [3] extends this direction by decomposing the task into three independent reasoning modules – a personal spatio-temporal memory module, a world-knowledge generator that models the influence of urban structure, and a collective-knowledge extractor – before aggregating the results in a final reasoning step. This approach outperforms earlier LLM baselines and exhibits lower geographical bias when evaluated across multiple cities.

However, a systematic evaluation by CityBench [6], covering 30 LLMs/VLMs across 13 cities, shows that while state-of-the-art models achieve competitive performance on tasks requiring common sense and semantic understanding, they *fail* on tasks requiring specialized knowledge and strong numerical competence, most notably geospatial prediction and traffic-signal control. TrajMLLM [7] reports a similar limitation: LLMs struggle to process numerical data and perform arithmetic operations, which limits their performance when directly encoding raw trajectory coordinates.

2.3 Combining LLMs with quantitative models: a research gap

Some works have begun to combine the two directions above. NextLocLLM [8] builds a coordinate-regression framework that uses a *single LLM backbone* both to generate semantic embeddings for POIs (via textual descriptions of location types) and to directly regress the next coordinate, subsequently mapping the result to the top- k nearest candidates via a post-hoc retrieval step. TrajAgent [9] takes an *orchestration* approach: an LLM agent selects and optimizes an appropriate specialized model for each specific task/dataset within a unified execution environment (UniEnv), achieving improvements of 2.38%–69.91% over baselines across four tasks.

What these two directions have in common is that the LLM and the quantitative model are *fused at the architectural level or at the task-orchestration level*, rather than exploiting a design with *separate inference stages at prediction time*: a quantitative model produces calibrated-probability candidates, and an LLM agent fuses and explains based on those candidates. Such a design offers three advantages that no existing approach achieves simultaneously: (i) it does not require fine-tuning the LLM backbone, unlike NextLocLLM; (ii) it narrows the LLM’s search space down to top- k candidates instead of the entire geographic space, reducing inference cost; (iii) it clearly separates the quantitative and semantic roles, enabling independent evaluation of each component’s contribution via ablation studies.

2.4 Bayesian networks for trajectory prediction and the motivation for combining them with LLMs

Before deep learning became the dominant paradigm, Bayesian networks were a fairly well-studied tool for trajectory prediction. A representative work [12] proposes a dynamic Bayesian network that models a sequence of variables – location, day of week, and time of day – to predict a user’s next location. Subsequent comparative studies between Bayesian networks and deep learning methods (RNNs) on real taxi trajectory data highlight an important advantage of Bayesian networks: statistical inference can be *updated in real time* as new trajectory data arrives, whereas RNN-based approaches require time-consuming retraining whenever new data

is added; the computational cost of predicting the next location is also substantially lower with Bayesian networks compared to deep learning.

However, these classical Bayesian models can only handle variables directly observable from numerical data (location, time), with no mechanism for incorporating unstructured contextual knowledge (location type, area semantics, habits expressed in natural language) as evidence variables in the network. This is the second gap that our proposed method targets: using an LLM as a *structured evidence extractor* that converts unstructured semantic knowledge into quantitative observed variables for a Bayesian network, combining the explicit statistical calibration and incremental-update capability of classical Bayesian networks with the contextual understanding of modern LLMs. To the best of our knowledge, this combination of Bayesian networks with an LLM agent has not been explored in recent trajectory-prediction work.

3 Proposed Method

3.1 Problem formulation

Given a user u with trajectory history $H_u = \{(p_1, t_1), \dots, (p_n, t_n)\}$, where $p_i \in \mathbb{R}^2$ is a GPS coordinate (or POI index) and t_i is the corresponding timestamp, the goal is to predict the next location p_{n+1} at query time t_{n+1} . Unlike traditional approaches that predict a single point directly, we formulate the problem at three levels:

- **Level 1 – Quantitative candidate generation:** learn a function $f_\theta : H_u \rightarrow \{(c_1, q_1), \dots, (c_k, q_k)\}$ that returns the top- k location candidates c_i together with calibrated probabilities q_i .
- **Level 2 – Semantic evidence extraction:** learn a function $g_{\text{LLM}} : (c_i, \mathcal{W}, \mathcal{M}_u) \rightarrow (h_i, s_i)$ for each candidate c_i , where $\mathcal{W}$ is world knowledge about the area, $\mathcal{M}_u$ is user u 's personal memory, $h_i \in [0, 1]$ is a personal-habit fit score, and $s_i \in [0, 1]$ is an urban-context (semantic/world) fit score for candidate c_i .
- **Level 3 – Bayesian fusion:** use a Bayesian belief network to compute the posterior $P(L = c_i \mid Q, H, S)$ over the candidate set, where L is the latent variable representing the true next location, Q is the quantitative evidence from Level 1, and H and S are the semantic evidence from Level 2, yielding the final prediction $\hat{p}_{n+1} = \arg \max_{c_i} P(L = c_i \mid Q, H, S)$ and a faithful explanation e derived directly from the probability decomposition.

3.2 Overall architecture

Figure 1 illustrates the three-stage processing pipeline:

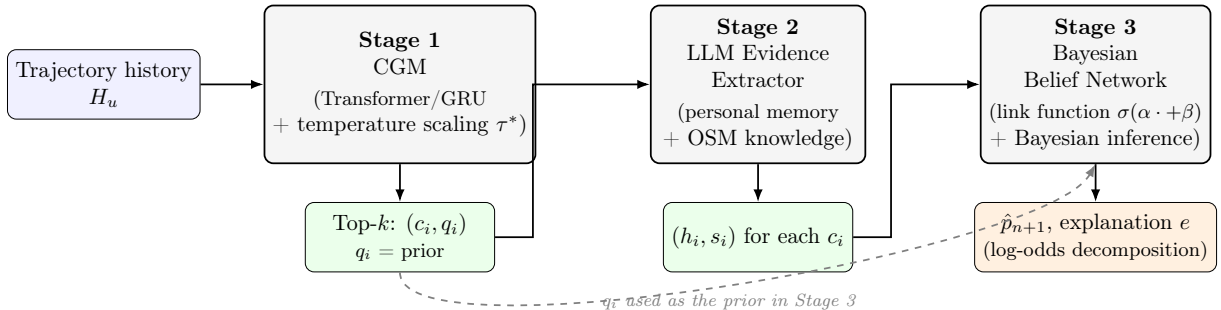


Figure 1: Overall three-stage architecture: (1) the quantitative CGM model generates candidates and the prior q_i ; (2) the LLM extracts semantic evidence (h_i, s_i) instead of directly re-ranking; (3) a Bayesian network fuses q_i, h_i, s_i into a posterior with a faithful explanation.

Details of each stage:

1. **Stage 1 (Candidate Generation Model – CGM)**: a lightweight quantitative model (a pretrained Transformer/GRU on trajectory data) takes the history H_u as input, produces a probability distribution over the location space, then takes the top- k candidates and applies *temperature scaling* to obtain probabilities q_i that accurately reflect the model’s true confidence. This probability q_i serves as the *prior* for Stage 3.
2. **Stage 2 (LLM Evidence Extractor – LEE)**: for each candidate c_i , the LLM receives two additional sources of knowledge – personal memory $\mathcal{M}_u$ (recurring movement patterns, frequent visit times) and world knowledge $\mathcal{W}$ (area function, distance, temporal context) – and performs chain-of-thought reasoning to output a *fit score* $h_i, s_i \in [0, 1]$ for each candidate, rather than directly producing a final ranking.
3. **Stage 3 (Bayesian Belief Network Fusion – BBN)**: the fit scores h_i, s_i are converted into likelihoods $P(H | L = c_i)$, $P(S | L = c_i)$ via a calibrated link function; the Bayesian network combines these likelihoods with the prior q_i to compute the final posterior, which is used both for ranking and for explanation.

3.3 Stage 1: Candidate generation and probability calibration

The CGM can be any specialized deep learning architecture for next-location prediction (e.g., GETNext or a simple autoregressive Transformer), trained with the standard log-likelihood maximization objective:

$$\mathcal{L}_{\text{CGM}} = - \sum_u \sum_{i=1}^n \log P_{\theta}(p_{i+1} | p_1, \dots, p_i). \quad (1)$$

Since classification models typically produce overconfident probabilities, we apply temperature scaling on the validation set:

$$q_i = \frac{\exp(z_i/\tau)}{\sum_{j=1}^{|\mathcal{C}|} \exp(z_j/\tau)}, \quad \tau^* = \arg \min_{\tau} \text{ECE}(\tau), \quad (2)$$

where z_i is the logit for candidate i , $\mathcal{C}$ is the entire location space, and τ^* is chosen to minimize the Expected Calibration Error (ECE) on the validation set. The top- k candidates with the highest q_i are passed on to Stage 2. This set of probabilities $\{q_i\}_{i=1}^k$ serves as the prior $P(L = c_i) = q_i$ for the Bayesian network in Stage 3.

3.4 Stage 2: Semantic evidence extraction with an LLM agent

The LLM agent inherits and adapts two reasoning modules from AgentMove [3], but *does not* directly output a final ranking as in the original design – instead, for each candidate c_i , the LLM outputs two scores that serve as input evidence for the Bayesian network in Stage 3:

- **Personal memory $\mathcal{M}_u \rightarrow$ score h_i** : instead of filtering purely by frequency (lossy counting) as in the original design, we add embedding-based retrieval: movement-history segments are encoded into vectors, and the top- m segments most similar to the current context are retrieved and included in the prompt. The LLM assesses the fit between c_i and the retrieved segments and returns $h_i \in [0, 1]$.
- **World knowledge $\mathcal{W} \rightarrow$ score s_i** : instead of relying solely on the LLM’s internal knowledge, we additionally retrieve real-world information about each candidate c_i (POI type, area density) from OpenStreetMap, injected into the prompt in a retrieval-augmented generation (RAG) fashion. The LLM assesses contextual fit (time, location type, distance) and returns $s_i \in [0, 1]$.

The LLM prompt follows the general template below (abridged), requesting scores rather than a direct ranking:

The user's movement history is: $\{\mathcal{M}_u\}$.
Consider candidate location c_i with context: $\{\mathcal{W}(c_i)\}$.
Reason step by step (chain-of-thought), then output a score from 0 to 1 for:
(a) fit with personal movement habits (h_i);
(b) fit with the current urban/temporal context (s_i).
Return only the two scores together with a brief rationale for each.

Requiring the LLM to return *per-candidate scores* rather than a *global ranking* makes the reasoning task simpler (independent/binary-style assessment rather than sorting an entire list), while producing an output format that is directly compatible with the Bayesian network in Stage 3. Because the input space has already been narrowed down to k candidates (typically $k \leq 10$) rather than the entire geographic space of the city, the number of tokens the LLM must process is substantially reduced compared to LLM-only approaches such as the original AgentMove.

3.5 Stage 3: Fusion via a Bayesian Belief Network

This is the central component that replaces "free-text LLM re-ranking" with an evidence-fusion mechanism that has explicit probabilistic semantics.

Network structure. We model the problem with a naive Bayesian network (Figure 2): a latent (query) variable L representing the true next location, and three observed child variables of L : Q (quantitative evidence from the CGM), H (personal-habit-fit evidence), and S (urban-context-fit evidence). The DAG structure is $L \rightarrow Q$, $L \rightarrow H$, $L \rightarrow S$, with the assumption that Q, H, S are conditionally independent given L – a reasonable simplifying assumption, since these three evidence sources originate from three independent data-generating mechanisms (an offline-trained quantitative model, personal history, and external urban knowledge).

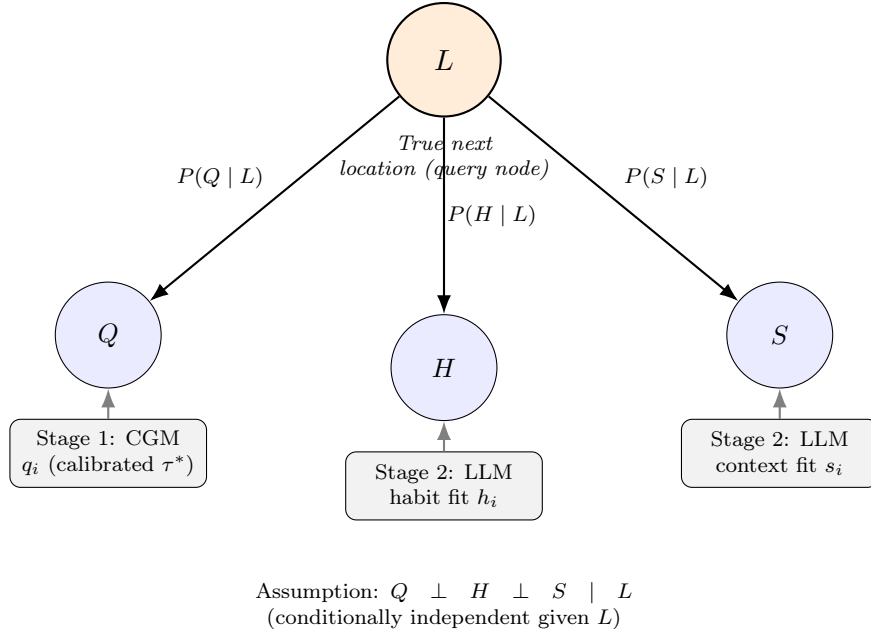


Figure 2: DAG structure of the naive Bayesian network used in Stage 3. The latent node L (true next location) generates the three observed evidence variables Q, H, S conditionally independently. The posterior $P(L \mid Q, H, S)$ is computed via Equation (4), with Q serving both as evidence and as the source of the prior $q_i = P(L = c_i)$.

Constructing likelihoods from LLM scores. We do not treat $h_i, s_i \in [0, 1]$ directly as probabilities; instead, we map them through a link function with calibration parameters (α, β) learned on the validation set (Platt scaling):

$$P(H = h_i \mid L = c_i) = \sigma(\alpha_H \cdot h_i + \beta_H), \quad P(S = s_i \mid L = c_i) = \sigma(\alpha_S \cdot s_i + \beta_S), \quad (3)$$

where $\sigma(\cdot)$ is the sigmoid function, and $(\alpha_H, \beta_H), (\alpha_S, \beta_S)$ are estimated via logistic regression minimizing the negative log-likelihood on the validation set, ensuring that the LLM’s subjective scores are converted into probabilities calibrated against real outcomes rather than being trusted blindly.

Bayesian inference. Given the prior $P(L = c_i) = q_i$ from Stage 1, the posterior over the k candidates is computed via Bayes’ rule under the conditional independence assumption:

$$P(L = c_i \mid Q, H, S) = \frac{q_i \cdot P(H = h_i \mid L = c_i) \cdot P(S = s_i \mid L = c_i)}{\sum_{j=1}^k q_j \cdot P(H = h_j \mid L = c_j) \cdot P(S = s_j \mid L = c_j)}. \quad (4)$$

The final prediction is $\hat{p}_{n+1} = \arg \max_{c_i} P(L = c_i \mid Q, H, S)$.

Faithful explanation. Because Equation (4) is an explicit computation, the contribution of each evidence source can be decomposed in log-odds form:

$$\log P(L = c_i \mid Q, H, S) = \underbrace{\log q_i}_{\text{quantitative contribution}} + \underbrace{\log P(H = h_i \mid L = c_i)}_{\text{personal-habit contribution}} + \underbrace{\log P(S = s_i \mid L = c_i)}_{\text{urban-context contribution}} - \log Z, \quad (5)$$

where Z is the normalization constant. The explanation e is generated by directly reporting these three terms (the candidate was chosen because it has a high/low q_i , a high/low h_i , a high/low s_i) – unlike LLM-only approaches, where the explanation is free-form text that is not guaranteed to reflect the computation that actually produced the result. This is the key distinction between BBN-fusion and LLM re-ranking: here, the explanation *is* the computation, not a post-hoc rationalization of it.

Incremental updating. Because of the naive-Bayes network structure, when a new evidence source becomes available (e.g., a signal from another sensor, or user feedback) at the next query, one only needs to multiply an additional likelihood factor into Equation (4), without retraining either the CGM or the LLM – directly inheriting the "real-time statistical-inference update" advantage of classical Bayesian networks for trajectory data [12], and overcoming the retraining requirement of purely deep-learning-based approaches.

3.6 Training and inference pipeline

Stage 1 is trained independently and offline, with no dependency on the LLM. Stage 2 operates in a zero-shot/few-shot manner via prompting, requiring no LLM fine-tuning, allowing flexible substitution among backbone LLMs (commercial or open-source, including small LoRA-tuned models). The calibration parameters of Stage 3 (τ^* for the CGM and (α, β) for the link functions) are estimated once on the validation set and held fixed at inference time. The full inference pipeline is summarized in Algorithm 1.

4 Experiments

This section presents the experimental design intended to answer the following research questions:

Algorithm 1 Three-stage inference for next-location prediction

Require: Trajectory history H_u , trained CGM f_θ , LLM agent g_{LLM} , number of candidates k , calibration parameters $\tau^*, (\alpha_H, \beta_H), (\alpha_S, \beta_S)$

Ensure: Prediction $\hat{p}_{n+1}$, explanation e

```
1:  $\{z_1, \dots, z_{|C|}\} \leftarrow f_\theta(H_u)$  ▷ Stage 1: generate logits
2:  $\{(c_1, q_1), \dots, (c_k, q_k)\} \leftarrow \text{TopK}(\text{softmax}(z/\tau^*), k)$  ▷ Calibrate + take top- $k$ ;  $q_i$  is the prior
3:  $\mathcal{M}_u \leftarrow \text{RetrievePersonalMemory}(H_u)$ 
4: for  $i = 1, \dots, k$  do
5:    $\mathcal{W}(c_i) \leftarrow \text{RetrieveWorldKnowledge}(c_i)$  ▷ OSM retrieval
6:    $(h_i, s_i) \leftarrow g_{\text{LLM}}(c_i, \mathcal{M}_u, \mathcal{W}(c_i))$  ▷ Stage 2: extract evidence
7:    $P(H=h_i \mid L=c_i) \leftarrow \sigma(\alpha_H h_i + \beta_H); \quad P(S=s_i \mid L=c_i) \leftarrow \sigma(\alpha_S s_i + \beta_S)$ 
8: end for
9:  $P(L=c_i \mid Q, H, S) \leftarrow \text{Equation (4)} \quad \forall i$  ▷ Stage 3: Bayesian inference
10:  $\hat{p}_{n+1} \leftarrow \arg \max_{c_i} P(L=c_i \mid Q, H, S)$ 
11:  $e \leftarrow \text{log-odds decomposition of } \hat{p}_{n+1}$  ▷ Faithful explanation
12: return  $\hat{p}_{n+1}, e$ 
```

RQ1 Does the proposed three-stage framework improve next-location prediction accuracy over purely quantitative baselines and LLM-only baselines?

RQ2 How large is the individual contribution of each component (probability calibration, embedding-based personal memory, OSM-retrieved world knowledge, and *especially the Bayesian fusion layer versus free-text re-ranking*)?

RQ3 Does the proposed framework reduce inference cost (number of tokens, latency) compared to LLM-only approaches, while maintaining or improving accuracy?

RQ4 Are the probabilities produced by the model (after fusion) better calibrated, and does the model generalize well across cities and across different backbone LLMs?

4.1 Datasets

Experiments are conducted on the following public trajectory datasets, following the setup of AgentMove [3] to ensure direct comparability:

- **ISP-Shanghai (WWW2019)**: real GPS trajectory data from a telecom carrier, recording continuous user locations in Shanghai.
- **Foursquare TIST2015**: check-in data from 12 global cities (Tokyo, New York, London, Paris, Sydney, Nairobi, Cape Town, Mumbai, San Francisco, São Paulo, Moscow, Beijing), enabling evaluation of cross-region generalization.
- **YJMob100K (extension)**: a larger-scale trajectory dataset used to test the scalability of the proposed framework beyond the original scope of AgentMove’s experiments.

We follow AgentMove’s trajectory-split protocol. Evaluation trajectories with fewer than four stays are excluded. For ISP-Shanghai, each user’s chronologically ordered trajectories are divided into 40% training, 10% validation, and 50% test partitions; the matched paper protocol then selects 521 validation queries and one test session for each of 200 users. Detailed statistics after preprocessing are reported in Table 1.

4.2 Baselines

We compare against four groups of baselines to clarify the contribution of each component of the proposed framework:

Table 1: Dataset statistics after preprocessing

Dataset	#Users	#Trajectories	#Locations	Avg. length	#Cities
ISP-Shanghai	1,230	6,050	3,536	4.5664	1
Foursquare TIST2015	22,946	306,324	236,666	7.3606	12
YJMob100K					

Group 1 – Purely quantitative models (no LLM).

- **Markov / Bi-gram**: a first-order state-transition probability model, used as a lower bound.
- **GETNext**: a Transformer combined with a trajectory flow map for next-POI recommendation.
- **CGM (Stage 1 only)**: the same candidate-generation model used in the proposed framework, evaluated in isolation by taking only the top-1 prediction without Bayesian fusion – used to measure the marginal contribution of Stages 2–3.

Group 2 – LLM-only (no underlying quantitative model).

- **LLM-Mob** [4]: direct LLM prompting to predict the next location from raw history.
- **LLM-ZS** [5]: a zero-shot variant with no illustrative examples.
- **AgentMove (original)** [3]: the three-module agentic framework (personal memory, world knowledge, collective knowledge), without a quantitative candidate-generation step.

Group 3 – Existing hybrid approaches.

- **NextLocLLM** [8]: coordinate regression using a single LLM backbone.
- **TrajAgent** [9]: an LLM orchestrating the selection of specialized models.

Group 4 – Classical Bayesian networks (no LLM).

- **Dynamic Bayesian Network (DBN)** [12]: a dynamic Bayesian network modeling the (location, day-of-week, time-of-day) sequence as in the original paper, with no LLM-derived semantic evidence – used to measure the true contribution of adding LLM-derived evidence to the Bayesian network.

Proposed method and ablation variants.

- **Ours (full)**: CGM (prior) + LLM evidence extraction (h_i, s_i) + Bayesian network fusion with a calibrated link function.
- **Ours w/o Calibration**: remove the temperature-scaling step in Stage 1 (uncalibrated prior q_i).
- **Ours w/o World Knowledge**: remove OSM retrieval, relying only on the LLM’s internal knowledge to generate s_i .
- **Ours w/o Personal Memory (embedding)**: revert to frequency-based filtering (lossy counting), as in the original AgentMove, to generate h_i .

- **Ours w/o BBN (raw LLM reranking)**: replace Stage 3 with direct free-text LLM re-ranking (as in the earlier iteration of the design), without explicit Bayesian fusion – the most important variant for isolating the specific contribution of the BBN layer.
- **Ours w/o Link Calibration**: use h_i, s_i directly as likelihoods (skipping Platt scaling (α, β)), to measure the value of the link-function calibration step.
- **Ours (Top-1 only)**: Stages 2–3 receive only a single candidate (to test whether extending to top- k is actually necessary).

4.3 Evaluation metrics

- **Acc@ k** ($k \in \{1, 5, 10\}$): the fraction of predictions for which the correct location appears within the final top- k result.
- **MRR (Mean Reciprocal Rank)**: measures the overall ranking quality of the candidate list after fusion.
- **ECE (Expected Calibration Error)**: measures how well predicted probabilities match empirical accuracy, evaluated separately for Stage 1 (before calibration), Stage 1 (after calibration), and the final Stage 3 output.
- **Inference cost**: average number of tokens per query (input + output) and average latency (seconds/query), compared between Ours and the LLM-only baselines.
- **Geographical bias**: the variance of Acc@1 across cities in Foursquare TIST2015, following AgentMove’s measurement protocol, to assess cross-region generalization.
- **Explanation faithfulness**: for Ours (full), we compare the final prediction under the original log-odds decomposition against predictions obtained after deliberately ablating each term ($\log q_i$, $\log P(H|L)$, $\log P(S|L)$) and checking whether the prediction changes in a manner consistent with the reported log-odds (a *counterfactual consistency check*); for the **Ours w/o BBN** variant, we measure the consistency between the LLM’s stated textual rationale and the actual ranking outcome, via human/LLM-judge evaluation on a small sample of size $-\text{[sample size]}$.

4.4 Implementation details

- **CGM model (Stage 1)**: a GRU candidate generator with 64-dimensional POI embeddings, 32-dimensional user embeddings, 16-dimensional time embeddings, and hidden dimension 128. It is trained with AdamW at learning rate 10^{-3} , batch size 64, for up to 10 epochs; the checkpoint maximizing validation Acc@1 plus Recall@10 is retained (epoch 7 for the reported run).
- **Temperature scaling**: τ^* minimizes validation negative log-likelihood through a deterministic golden-section search over $\tau \in [e^{-4}, e^4]$; the fitted value is $\tau^* = 2.6370$.
- **Number of candidates k** : default $k = 10$, with the planned sensitivity study using $k \in \{1, 3, 5, 10\}$ in Section 4.9.
- **Backbone LLM (Stage 2)**: the currently reported Shanghai experiments use Qwen2:7b through the local Ollama OpenAI-compatible endpoint. A second backbone remains required for the cross-backbone analysis.
- **World-knowledge retrieval**: the Nominatim/OpenStreetMap service (self-hosted or via the public API) to obtain POI type and four-level address information for each candidate.

- **Embedding-based personal memory:** a deterministic 132-dimensional structured encoder hashes category and POI identity features and appends sinusoidal hour/weekday features. Cosine similarity retrieves the top- m segments, with $m = 5$.
- **BBN link-function calibration:** $(\alpha_H, \beta_H) = (2.3891, -3.4452)$ and $(\alpha_S, \beta_S) = (2.2942, -3.4072)$ are fitted by an in-repository Newton solver for one-dimensional logistic regression on 521 validation queries (5,210 candidate examples before invalid-output filtering). The resulting posterior odds are converted to prevalence-corrected likelihood ratios capped at 20.
- **Hardware:** one MacBook Pro with an Apple M4 Pro (12-core CPU, 16-core integrated GPU) and 24 GB unified memory. Qwen2:7b is served locally by Ollama, so the reported runs incur no external API charge. Stage-1 wall-clock training time was not logged and is therefore not reported.

4.5 Main results (RQ1)

Table 2 reports Acc@1, Acc@5, Acc@10, and MRR on both datasets. The TIST2015 Markov/Bi-gram, CGM, no-OSM Hybrid, LLM-Mob, and LLM-ZS entries are unweighted macro averages over all 12 cities and 2,382 matched test queries. Markov/Bi-gram ranks the full per-city candidate space using smoothed first-order transition logits. The current AgentMove entry is an explicitly marked partial macro average over five completed cities and 506 queries; the 29 incomplete Paris predictions are excluded. The Hybrid and AgentMove runs have no OSM enrichment. Because the matched LLM-only prompts request only five predictions, their Acc@10 cells are left blank. The TIST2015 LLM-Mob reproduction omits visit duration because this field is unavailable in the processed JSONL.

Table 2: Main results compared against baselines. TIST2015 values marked $\dagger$ are complete 12-city no-OSM ablation results. Values marked $\ddagger$ are interim AgentMove no-OSM macro averages over 5/12 cities (506 queries). Blank cells denote unavailable experiments or metrics.

Group	Method	ISP-Shanghai				Foursquare TIST2015 (12-city macro avg.)			
		Acc@1	Acc@5	Acc@10	MRR	Acc@1	Acc@5	Acc@10	MRR
Quantitative	Markov/Bi-gram	0.0940	0.1855	0.1997	0.1374	0.1121	0.2103	0.2423	0.1585
	GETNext								
	CGM (Stage 1 only)	0.1950	0.4350	0.4850	0.3005	0.1796	0.3359	0.3893	0.2466
LLM-only	LLM-Mob (Qwen2:7b)	0.1200	0.3950		0.2200	0.1723	0.3664		0.2455
	LLM-ZS (Qwen2:7b, history-only)	0.1800	0.4050		0.2572	0.0988	0.3055		0.1743
	AgentMove (Qwen2:7b; no-OSM partial)	0.1650	0.4100		0.2586	0.1191 $\ddagger$	0.3422 $\ddagger$		0.1968 $\ddagger$
Existing hybrids	NextLocLLM								
	TrajAgent								
Classical Bayes	DBN [12]								
Proposed	Ours (Shanghai full; TIST no-OSM)	0.1900	0.4250	0.4800	0.2939	0.1800 $\dagger$	0.3342 $\dagger$	0.3848 $\dagger$	0.2476 $\dagger$

Discussion. On the matched 200-user ISP-Shanghai test set, Ours improves over the reproduced LLM-based baselines but remains slightly below Stage 1. Across all 12 TIST2015 cities, the no-OSM Hybrid obtains macro Acc@1, Acc@5, Acc@10, and MRR of 0.1800, 0.3342, 0.3848, and 0.2476, respectively, compared with 0.1796, 0.3359, 0.3893, and 0.2466 for Stage 1. The Markov/Bi-gram baseline obtains 0.1121, 0.2103, 0.2423, and 0.1585, confirming that the learned Stage-1 model provides a substantial gain over a first-order transition model. Fusion produces only a marginal Acc@1/MRR gain over Stage 1 while slightly reducing Acc@5/Acc@10. LLM-Mob reaches 0.1723 Acc@1, 0.3664 Acc@5, and 0.2455 MRR, whereas LLM-ZS reaches 0.0988, 0.3055, and 0.1743. The interim AgentMove no-OSM result over five completed cities is 0.1191 Acc@1, 0.3422 Acc@5, and 0.1968 MRR; it is not directly comparable with the completed 12-city rows until the remaining cities finish. OSM-enriched metadata remains required before claiming the full TIST2015 model.

4.6 Ablation study (RQ2)

Table 3 reports ablation results on the ISP-Shanghai dataset, isolating the contribution of each component.

Table 3: Ablation study on the AgentMove-compatible 200-user ISP-Shanghai test set. Blank cells denote exact manuscript variants not yet available.

Variant	Acc@1	Acc@5	MRR	ECE (final output)
Ours (full)	0.1900	0.4250	0.2939	0.0971
w/o Calibration (CGM)	0.1900	0.4450	0.2992	0.5438
w/o World Knowledge (OSM)	0.2000	0.4400	0.2998	0.1046
w/o Personal Memory (embedding)	0.1950	0.4400	0.2991	0.0893
w/o BBN (raw LLM reranking)	0.1800	0.4350	0.2890	
w/o Link Calibration (α, β)	0.1850	0.4200	0.2879	0.0890
Top-1 only (no Stages 2–3)	0.1950	0.4350	0.3005	0.0949

The **w/o BBN (raw LLM reranking)** row is the paper’s single most important comparison: it isolates precisely the added value of replacing "free-text LLM re-ranking" with "evidence fusion via an explicit Bayesian network," while all other components (CGM, personal memory, world knowledge) remain unchanged.

Discussion. Likelihood-ratio link calibration improves Acc@1 from 0.185 to 0.190 and Acc@5 from 0.420 to 0.425, although its ECE is slightly higher than the uncalibrated-link variant (0.0971 versus 0.0890). Removing CGM temperature scaling leaves Acc@1 unchanged and raises Acc@5 to 0.445, but severely degrades ECE from 0.0971 to 0.5438, demonstrating a strong accuracy–calibration trade-off. Replacing the BBN with direct free-text reranking reduces Acc@1 from 0.190 to 0.180 and MRR from 0.2939 to 0.2890, while increasing Acc@5 from 0.425 to 0.435. The paired Acc@1 difference has a 95% bootstrap interval of $[-0.015, 0.035]$ and is therefore not statistically significant. Moreover, 14% of free-text responses are malformed or incomplete and require deterministic Stage-1 completion, whereas BBN inference always produces a normalized posterior. Removing OSM world knowledge increases Acc@1 from 0.190 to 0.200, Acc@5 from 0.425 to 0.440, and MRR from 0.2939 to 0.2998, while ECE worsens from 0.0971 to 0.1046. The paired full-minus-no-OSM Acc@1 difference is -0.010 with a 95% bootstrap interval of $[-0.030, 0.010]$, so the difference is not statistically significant. Replacing embedding retrieval with frequency-based personal memory likewise increases Acc@1 from 0.190 to 0.195, Acc@5 from 0.425 to 0.440, and MRR from 0.2939 to 0.2991, while improving ECE from 0.0971 to 0.0893. However, the paired full-minus-frequency Acc@1 difference is only -0.005 with a 95% bootstrap interval of $[-0.025, 0.015]$. Thus, neither OSM nor embedding retrieval provides a statistically supported accuracy gain on this Shanghai test subset; these negative results motivate improving their representations rather than claiming unsupported component benefits.

4.7 Inference cost analysis (RQ3)

Table 4 compares inference cost between the proposed method and LLM-only baselines. The reported LLM-ZS run uses Qwen2:7b locally through Ollama on the AgentMove-compatible 200-user ISP-Shanghai test set.

Discussion. Compact extraction reduces the Hybrid pipeline from the original AgentMove’s 3.0 calls to 1.015 LLM calls per query. Ours uses approximately 1,069.03 input-plus-output tokens and 8.78 seconds per query, compared with 3,086.85 tokens and 26.38 seconds for AgentMove original, corresponding to reductions of 65.4% in tokens and 66.7% in measured LLM latency

Table 4: Comparison of inference cost (average per query) on the same AgentMove-compatible 200-user ISP-Shanghai test set. Token counts use Ollama usage when exposed and a text-length estimate otherwise; latency reports measured LLM-call time. Blank cells denote experiments not yet available.

Method	#Input tokens	#Output tokens	Latency (s)
LLM-Mob (Qwen2:7b)	685.96	137.22	3.43
LLM-ZS (Qwen2:7b)	361.77	49.31	8.10
AgentMove original (Qwen2:7b)	2099.98	986.87	26.38
Ours (full)	902.70	166.33	8.78

while increasing Acc@1 by 0.025. Compared with the simpler LLM-only baselines, LLM-ZS uses 411.08 tokens and 8.10 seconds, while LLM-Mob uses 823.18 tokens and 3.43 seconds; Hybrid therefore does not reduce cost relative to these one-call prompts. When the BBN is replaced by exact free-text Stage-3 reranking, that reranker adds 890.56 estimated tokens and 5.26 seconds per query; BBN inference itself is effectively instantaneous after structured evidence extraction. RQ3 is therefore supported relative to the original multi-call AgentMove and free-text fusion, but not as a universal cost reduction over simpler LLM-only methods.

4.8 Calibration and generalization analysis (RQ4)

Figure 3 illustrates the reliability of Stage 1 before/after temperature scaling and of the final output after Bayesian fusion.

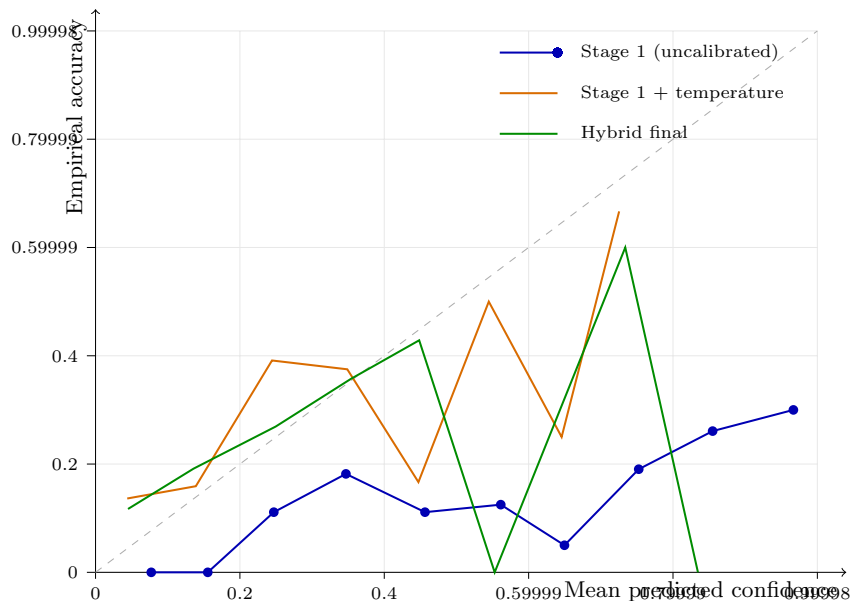


Figure 3: Reliability diagram on the matched 200-query ISP-Shanghai test set. Points aggregate predictions into ten equal-width confidence bins; empty bins are omitted.

Table 5 reports per-city Acc@1 on Foursquare TIST2015, together with the cross-city variance (geographical bias), for different backbone LLMs.

Discussion. Temperature scaling substantially reduces Stage-1 ECE from 0.5100 to 0.0949 without changing Acc@1 (0.1950). Bayesian fusion yields ECE 0.0971 and Acc@1 0.1900, preserving most of the calibration improvement but not improving accuracy over calibrated Stage

Table 5: Acc@1 by city and by backbone LLM (Foursquare TIST2015). The Qwen2:7b row is the complete 12-city no-OSM ablation; variance is the population variance across cities.

Backbone LLM	Tokyo	NY	London	Paris	Sydney	Nairobi	CapeTown	Mumbai	SF	SaoPaulo	Moscow	Beijing	Variance
Qwen2:7b (no-OSM)	0.1200	0.1950	0.1500	0.1300	0.1100	0.1250	0.3297	0.1500	0.2100	0.2750	0.2250	0.1400	0.004333
Backbone B (pending)													

1. Across the complete 12-city no-OSM TIST2015 run, Acc@1 ranges from 0.1100 to 0.3297 and has a population variance of 0.004333. This quantifies the current geographical variation, but it remains an ablation result until OSM enrichment satisfies the publication gate.

4.9 Sensitivity to the number of candidates k

A planned sensitivity analysis will examine Acc@1 and inference cost as $k \in \{1, 3, 5, 10\}$ varies, in order to identify a reasonable trade-off point between accuracy and token cost.

4.10 Qualitative analysis (case study)

We present $-[\text{number}]$ illustrative examples in which Stage 3 changes the ranking relative to Stage 1’s raw output, together with the LLM-generated rationale, to visually demonstrate the value of the semantic fusion step (e.g., a case where the CGM predicts incorrectly due to missing temporal/event context, but the fused prediction is corrected via world knowledge).

`-[Insert a real example: movement history, original top- $k$  candidates, the result after fusion, and the explanation  $e$  generated by the LLM.]`

References

- [1] W. Zhang, J. Han, Z. Xu, H. Ni, H. Liu, H. Xiong. Urban Foundation Models: A Survey. *Proc. KDD*, 2024.
- [2] W. Chen, Y. Liang, Y. Zhu, Y. Chang, K. Luo, H. Wen, L. Li, Y. Yu, Q. Wen, C. Chen, et al. Deep Learning for Trajectory Data Management and Mining: A Survey and Beyond. *arXiv preprint arXiv:2403.14151*, 2024.
- [3] J. Feng, Y. Du, J. Zhao, Y. Li. AgentMove: A Large Language Model based Agentic Framework for Zero-shot Next Location Prediction. *Proc. NAACL*, 2025.
- [4] X. Wang, et al. Where Would I Go Next? Large Language Models as Human Mobility Predictors. *arXiv preprint arXiv:2308.15197*, 2023.
- [5] C. Beneduce, B. Lepri, M. Luca. Large Language Models are Zero-Shot Next Location Predictors. *IEEE Access*, 2025.
- [6] J. Feng, et al. CityBench: Evaluating the Capabilities of Large Language Models for Urban Tasks. *Proc. KDD*, 2025.
- [7] Liu, et al. Traj-MLLM: Can Multimodal Large Language Models Reform Trajectory Data Mining? *Proc. KDD*, 2025.
- [8] Anonymous. NextLocLLM: Location Semantics Modeling and Coordinate-Based Next Location Prediction with LLMs. *arXiv preprint arXiv:2410.09129*, 2024.
- [9] Y. Du, J. Feng, J. Zhao, Y. Li. TrajAgent: An Agent Framework for Unified Trajectory Modelling. *arXiv preprint arXiv:2410.20445*, 2024.

- [10] Y. Zhu, J. J. Yu, X. Zhao, X. Wei, Y. Liang. UniTraj: Universal Human Trajectory Modeling from Billion-Scale Worldwide Traces. *arXiv preprint arXiv:2411.03859*, 2024.
- [11] X. Wu, H. He, Y. Wang, Q. Wang. Pretrained Mobility Transformer: A Foundation Model for Human Mobility. *arXiv preprint arXiv:2406.02578*, 2024.
- [12] A. Asahara, K. Maruyama, A. Sato, K. Seto. Pedestrian-movement prediction based on mixed Markov-chain model. *Proc. ACM SIGSPATIAL GIS*, 2011. (See also related work on dynamic Bayesian networks for trajectory prediction, e.g., “Analysis and prediction of trajectories using Bayesian network”.)